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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Professional Training I

"MoodMeal" — A Mood-Based Recipe & Nutrition Recommender

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AGENDA

- Certificate
- Abstract
- Introduction
- Objective
- System Architecture / Ideation Map
- Module Implementation
- Results and Discussions
- Conclusion
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ABSTRACT

Most recipe and diet-tracking applications recommend meals based only on available ingredients or dietary restrictions, overlooking the user's mood and situational context. MoodMeal is a desktop application, developed using Java (Swing/JavaFX) and MySQL, that recommends recipes based on the user's current mood, available pantry ingredients, and basic nutritional goals.

Recipes are tagged with mood-suitability scores, nutrition values, and ingredients, allowing users to receive filtered suggestions displayed as visual cards with prep time and a "comfort vs. energy" rating. The system also tracks mood-to-meal history, revealing simple eating patterns over time through basic charts. The project applies core Java concepts, JDBC-based MySQL CRUD operations, and GUI design to an underexplored intersection of nutrition and emotional wellbeing.

INTRODUCTION

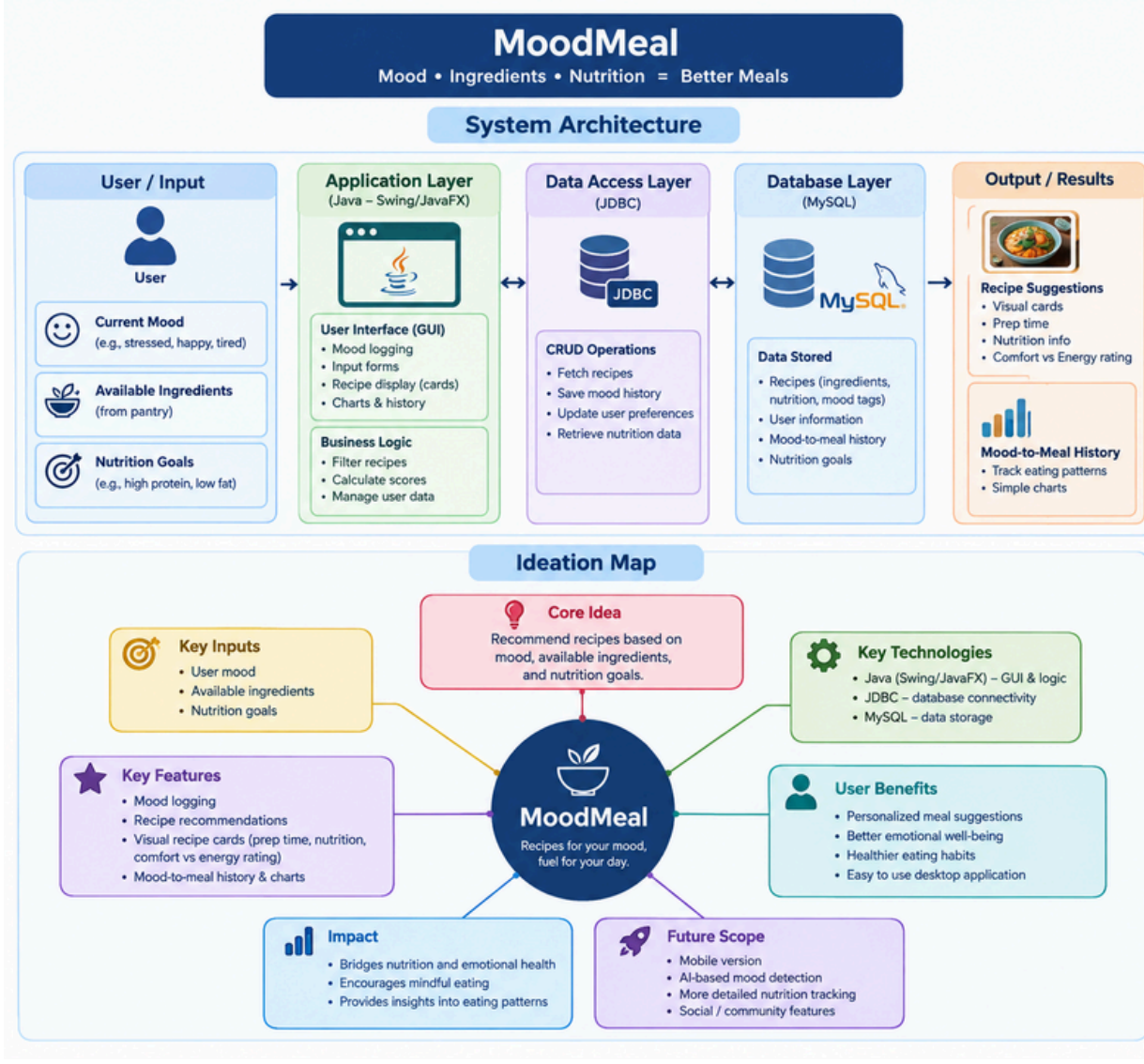
Diet and recipe recommendation applications have become increasingly common, but most of them operate on a narrow set of inputs — typically the ingredients a user has on hand or a fixed dietary restriction such as vegetarian, low-carb, or gluten-free. While useful, this approach treats meal selection as a purely logistical problem, ignoring the psychological and situational factors that actually influence what people choose to eat on a given day. In reality, food choices are rarely made in a vacuum; they are shaped by how a person is feeling in that moment.

Emotional eating is a well-documented behavioral pattern, where stress, fatigue, boredom, or celebration often drive people toward specific types of food, such as comfort food during stressful periods or lighter meals when energy is low. Existing nutrition and recipe platforms largely overlook this connection, leaving a gap between how people actually decide what to eat and how technology tries to help them decide. MoodMeal is designed to address this gap directly, bridging nutrition technology with emotional wellbeing by incorporating the user's mood as a core factor in generating recipe recommendations, alongside available ingredients and nutritional goals.

OBJECTIVE

The objective of this project is to build a desktop application that recommends recipes based on mood, available ingredients, and nutrition goals. The application allows users to log their daily mood and track eating patterns over time, while providing visually engaging recipe suggestions through cards showing prep time, nutrition information, and a comfort-versus-energy rating. It also demonstrates the practical integration of core Java, JDBC, and MySQL in a real-world application.

System Architecture / Ideation Map



MODULE IMPLEMENTATION

1. **User Module** — Handles user registration, login, and profile management; includes the mood logging interface where users record their current emotional state (e.g., stressed, low-energy, celebratory) on a daily basis.
2. **Recipe Module** — Manages the storage and organization of recipes, including mood-suitability tagging, nutritional values (calories, protein, sugar, etc.), and ingredient lists for each recipe.
3. **Recommendation Module** — Contains the core filtering logic that matches recipes to the user's current mood, available pantry ingredients, and specified nutritional goals, generating a ranked list of suitable suggestions.
4. **History & Analytics Module** — Tracks mood-to-meal history over time and generates simple visual patterns and charts (using JFreeChart) to help users understand trends, such as recurring comfort-food choices on stressed days.
5. **Database Module** — Manages JDBC connectivity between the application and the MySQL database, handling all CRUD (Create, Read, Update, Delete) operations for users, recipes, ingredients, and mood logs.

RESULTS AND DISCUSSIONS

The system successfully generates mood-based recipe recommendations from a sample dataset, with visual cards presenting recipe details clearly and intuitively. History tracking reveals meaningful patterns, such as a tendency toward comfort-food selection on stressed days. The discussion also considers system performance with larger datasets and the accuracy of mood-suitability tagging as areas for further refinement.

CONCLUSION

MoodMeal demonstrates a functional, mood-aware recipe recommendation system that successfully integrates Java GUI development with MySQL-based data persistence. It highlights an underexplored intersection between nutrition and emotional wellbeing. Future scope includes developing a mobile version, incorporating ML-based mood detection, and expanding the recipe database.

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THANK YOU

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